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EfficientNet-Based Transfer Learning Framework for Indian Cattle Breed Recognition

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Abstract*—The purpose of this research is to create a system that can predict the different breeds of cattle and buffalos such as 'Amritmahal', 'Ayrshire', 'Bargur', 'Dangi', 'Deoni', 'Gir', 'Hallikar', 'Hariana', 'Himachali Pahari', 'Kangayam', 'Kankrej', 'Kenkatha', 'Khariar', 'Khillari', 'Konkan Kapila', 'Kosali', 'Krishna_Valley', 'Ladakhi', 'Lakhimi', 'Malnad_gidda', 'Mewati', 'Nari', 'Nimari', 'Ongole', 'Poda Thirupu', 'Pulikulam', 'Punganur', 'Purnea', 'Rathi', 'Red kandhari', 'Red_Sindhi', 'Sahiwal', 'Shweta Kapila', 'Tharparkar', 'Umblachery', 'Vechur', 'bachaur', 'badri', 'bhelai', 'dagri', 'gangatari', 'gaolao', 'ghumsari', 'kherigarh', 'malvi', 'motu', 'nagori', 'ponwar', 'siri' and 'thutho'. There are multiple breeds of cow and bufflo in India which are essential for genetic conservation, dairy productivity and an efficient livestock management. The dataset consisted of images related to 50 different breeds. The method used in this research is a Deep Learning based classification to train the model by differentiating between the characteristics of each cattle. Later the model is used to predict the cattle breed present in an image using EfficientNet and Transfer Learning. During the training process a transfer learning strategy was used, based on the EfficientNetV2-S architecture which is already pretrained on the ImageNet dataset. Initially, the pretrained data is used as the feature extractor and a custom classification head is added to allow the network to adapt for cattle breed recognition. The model training is done using the Adam optimizer and cross-entropy is the loss function. Later to improve the generalization and reduce the risk overfitting, many techniques such as data augmentation, dropout and normalization are used during the model training process. The experimental results of this approach demonstrated that EfficientNet based transfer learning approach is very much effective in capturing the visual features such as structure of the horn, facial patterns, texture of the body and the color variations, this enables accurate breed classification

Index Terms*: Deep Learning, Image Classification, EfficientNet, Transfer Learning, Cattle Breed Classification, Cattle Breed Recognition, and Livestock Informatics

I. INTRODUCTION.

India has a wide variety of Cattle breeds that are adapted to different climatic conditions and are present in various geographical locations. They essential for dairy productivity, agriculture, livestock management, preservation of genetic biodiversity and also play a major role in rural economies. Many breeding programs, productivity optimization, disease control and the preservation of endangered species depend on accurate breed identification. Each cattle breed exhibits specific characteristics for instance, skin patterns may vary extensively and be strikingly attractive, body colors can be rare and captivating, and the shapes of their horns might be quite unique, that helps to differentiate one breed from another.

Therefore, the correct breed identification not only promotes various opportunities but also requires knowledge about the diverse breed of cattle and buffalos. In various breed recognition techniques manual identification is the most common practice. This requires expertise knowledge and is also error prone sometimes as few breeds might look similar with the visual appearance and there might be only a slight change in their characteristics.

This research addresses a common issue in the recognition of livestock breeds through the application of deep learning models like EfficientNet with transfer learning, this research successfully identifies different cattle breeds. However, one identifiable remark about the study is, the dataset is expandable. The images related to the new breeds can be placed in the placed in the folder named by the breed and can be moved to the dataset for training. After the training the models is saved and the recognised breed that can be predicted by the model is updated and the new breed is also predicted.

II. LITERATURE REVIEW.

Advances Research on breed identification of different buffalos and cattle using DenseNet121 using Transfer Learning is not widely conducted. Yet, numerous studies have been done in the field of breed identification and breed classification. Some of these methods have used traditional Machine Learning models and Deep Learning models. One study titled

[1] "A Comprehensive Analysis of Classic Machine Learning and Deep Learning Modelling for Breed Recognition of Bananas," focused on identifying banana breeds using traditional machine learning models and deep learning models. It showcases the effectiveness of using traditional machine learning models and deep learning models to identify the banana breed. This approach involves the banana image analysis to predict the breed. By performing the task of different models of machine learning and deep learning like Support Vector Machine confirms (SVM), Random Forest, K-Nearest Neighbours and Convolutional Neural Network (CNN). The study confirms that Deep Learning is suitable for image-based recognition.

The study [2] "A Vision-Based Cattle Recognition System Using TensorFlow for Livestock Water Intake Monitoring", focused on vision-based cattle recognition system by capturing the image of the animals when it approached near water and recognised individual cattle using Deep learning models like Convolutional Neural Network (CNN) and TensorFlow API for the object detection the water intake per cattle was monitored. This study confirms Convolutional Neural Network (CNN) is suitable for livestock image analysis.

The study titled [3] "Cattle Breed Detection and Categorization Using Image Processing and Machine Learning", addresses the

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method of automatically identifying the cattle breeds using images instead of manual inspection. The images belonging to different breeds were collected and image processing techniques were used to extract the feature used Machine Learning classifiers such as Support Vector Machine (SVM), Random Forest, K-Nearest Neighbours (KNN) and decision Trees. The study used a small dataset with limited breeds, with the images captured in controlled environment. By this the system achieves moderate accuracy, demonstrates image-based classification is possible, performs well on small dataset.

The study titled [4] "Contour Extraction of Individual Cattle from an Image Using Enhanced Mask R-CNN Instance Segmentation Method', addresses the problem of accurately separating individual cattle from complex background images. The system detects the cattle in the image, extracts precise contour of each cattle. This study mainly focused on preprocessing using images containing single and multiple cattle, complex backgrounds where Resnet-Convolutional Neural Network (R-CNN) was used as the model to classify the images and the model performed with the high segmentation accuracy and a clean contour extraction.

The study titled [5], "Cow Face Recognition with Noise Labels Using Vision Transformer," addresses the problem of recognizing individual cows using their face images when the dataset contains noisy data. Face images of 6 individual cows, total of 3556 images of cow faces where captured and some noise labels were introduced. Vision Transformer was used to split the images into patches and robust training was used to handle the noise and train the data. The model performs with high accuracy in identifying individual cows with a good performance with the noisy labels.

The study titled [6] "Dog Breed Detection Using YOLOv7 and WOA-CNN," addressed the problem of automatically detecting the dogs in the images and identifying their breeds. Standford Dog Dataset, with 120 breeds were used as the dataset. The deep learning model YOLOv7(You only look once) was used for the object classification, Convolutional neural network (CNN) for breed classification along with Whale optimization (WOA) Technique and deep learning frameworks such as TensorFlow were used to predict the breed of the of uploaded dog image. The model performed with high breed classification accuracy, real-time performance using the dog images.

The study titled [7] "Identifying the Optimal Deep Learning Based Image Recognition Technique for Dog Breed Detection," was done for various deep learning models like ResNet, Inception, Mobile Net for finding the best fit model for breed recognition tasks. The dog breed image dataset was used to train these models and their performances were compared to find the best model among them. Models were evaluated based on their accuracy, training time, computation complexity and memory usage. The all the model's accuracy and speed were checked. This study shows the importance of model comparison and selecting the best fi model.

The study titled [8] "Graph Synthesis for Pig Breed Classification from Muzzle Images", automatically classifies Pig breed using biometric features. High resolution close-up images of pigs were collected and the image preprocessing and texture analysis was done and then the extracted features were converted into nodes and edges for the graph synthesis. These nodes and edges were used to plot the graph and the breed was recognised based on the graph similarity. The model achieved good accuracy and gave an idea on using the visual features for breed detection.

The study titled [9] "Research on Multi-View Cattle Facial Recognition Algorithm Based on Feature Matching," recognized individual cattle using facial images captured from different angles. The facial features were extracted the features were matched across the view and recognised individual cattle. Techniques such as image processing, feature extraction, feature matching were used to extract the features from the image. The model performed with an improved recognition accuracy compared to the single-view systems.

The study titled [10] "Identification of cattle breed type by processing images using DL Models", was done to identify the breed of a cattle image with a small dataset using the deep learning model Convolutional Neural Network (CNN) that support automatic feature extraction, was trained using supervised learning and used python as the programming language. The model achieved a high accuracy and the image-based breed recognition using Convolutional Neural Network (CNN) was demonstrated.

The study [11] "Video-Based Analysis of Cattle Behaviours: Improved Classification Using FlowEQ Transform," was done to automatically analyse and classify cattle behaviour using video data instead of manual observation. The video recording of the cattle was done with the multiple cattle behaviours as walking, standing, laying down. A video-based feature extraction was done to monitor the behaviour of the cattle. This helped in the separation between the similar behaviour with an improved accuracy.

The study [12] "Automatic Cattle Identification Based on Fusion of Texture Features Extracted From Muzzle Images," was conducted to identify the cattle from their biometric features. The muzzle images of the cattle were used to extract the features of individual cattle. Later the images were processed and the texture features were extracted using single channel Local binary pattern (LBP), the classification and identification was done using machine learning models like K-Nearest Neighbours (KNN), Support Vector Machine (SVM). The system showed high identification accuracy in controlled conditions.

The study [13] "Automatic Cattle Identification based on Multi-Channel LBP on Muzzle Images," was conducted to identify the cattle from their biometric features. The muzzle images of the cattle were used to extract the features of individual cattle. Later the images were processed and the texture features were extracted using Multi-Channel Local binary pattern (LBP), the classification and identification was done using machine learning models like K-Nearest Neighbours (KNN), Support Vector Machine (SVM). High resolution images were used, the system showed high identification accuracy in controlled conditions compared to the single channel method.

The study [14] "Identification of Different Types of Buffalo Using YOLO Algorithm Based on Computer Vision," was done to identify the buffalo from the images using the You Only Look Once (YOLO) model and classifies them into different types. Images of different buffalo were collected and the model was trained using these images. Later the accepts the input image, detects and classifies the buffalo.

The study [15], "A hybrid deep learning based robust framework for cattle identification," was conducted to identify the individual cattle in real-world farm environments. Multiple type of cattle images including the muzzle images and face images was collected. Later these images went through the image processing, denoising, Convolutional Neural Network (CNN) was used for feature extraction. The model demonstrated high identification accuracy for controlled farm environment images.

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While the previous studies have made the studies in image recognition, breed detection, model selection. Study [7] requires enhanced accuracy. Study [8] and [9] need to refine the image capturing, rather than capturing the images in the controlled environment. Unlike these the study[10] utilized Convolutional Neural Network (CNN) on focused on identifying the cattle breeds. The novelty of this research lies in its use of a never-before-applied-method, EfficientNet focus on cattle and buffalo breed detection using their images, a subject not previously explored. The transfer learning aim to advance the recognition of various cattle and buffalo, particularly based on their distinct features and is is anticipated to elevate the precision of Cattle breed recognition significantly.

The difference between this study and the previous studies lies in three aspects, namely the object of research, the method used and the mode of deployment. The object of this research is Cattle. The method used in this research is EfficientNet and Transfer Learning, the mode of deployment is using streamlit framework, which is the latest method and has never been used before to detect cattle breed. This research contributes to the development of the system that is able to accurately detect the various types of cattle breeds. This research uses classification methods in pattern recognition, training a model to distinguish the different characteristics of each breed. This model is then used to identify cattle species in the image using the using the EfficientNet and Transfer Learning. The trained model is later saved and deployed into web application using the streamlit framework.

III. PROPOSED METHOD.

The research focuses on creating a system that can recognize different cattle and buffalo species using their images through a deep learning technique that incorporate the "EfficientNet and Transfer Learning" methods. The dataset consists of 8379 images belonging to various cattle breeds belonging to different regions across in India. These images have been captured using mobile phones and cameras. Capturing various views and instances where multiple cows may appear in a single image.

3.1 Processing Stage.

The flow diagram show in the Figure 2 is EfficientNet designed platform for cattle and buffalo breed recognition begins with the dataset upload in the .zip file format. This dataset is then extracted and organized.

The extracted dataset is passed on to the training using EfficientNet, the performance was evaluated and the model is saved for the further use by streamlit. It has five different stages. Dataset Upload, Dataset extraction, Pre-Trained EfficientNet, model training and then the model saving that allows the model to be reused. Let us understand this process with the help of a Block Diagram and in detail explanation about the model.

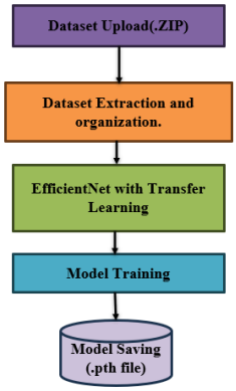

Figure 1. Block diagram of the proposed system.

3.1.1 Dataset Upload.

The dataset was sourced from multiple resources representing multiple cattle and buffalo breeds in India. The model utilized of over 8379 images, which was categorized in the different folders belonging to separate breeds, the folder was named by the breed name and the images inside the folder was named by the breed name followed by the extension.The model identifies different cattle breeds: 'Amritmahal', 'Ayrshire', 'Bargur', 'Dangi', 'Deoni', 'Gir', 'Hallikar', 'Hariana', 'Himachali Pahari', 'Kangayam', 'Kankrej', 'Kenkatha', 'Khariar', 'Khillari', 'Konkan Kapila', 'Kosali', 'Krishna_Valley', 'Ladakhi', 'Lakhimi', 'Malnad_gidda', 'Mewati', 'Nari', 'Nimari', 'Ongole', 'Poda Thirupu', 'Pulikulam', 'Punganur', 'Purnea', 'Rathi', 'Red kandhari', 'Red_Sindhi', 'Sahiwal', 'Shweta Kapila', 'Tharparkar', 'Umblachery', 'Vechur', 'bachaur', 'badri', 'bhelai', 'dagri', 'gangatari', 'gaolao', 'ghumsari', 'kherigarh', 'malvi', 'motu', 'nagori', 'ponwar', 'siri' and 'thutho'. Organizing the dataset in a structured folder format allows the deep learning framework to automatically assign labels during the training process.

3.1.2 Dataset Extraction and Organization.

The dataset is then extracted and the further image preprocessing takes place. The image is resized to 224x224.The image pixel values are normalized. Data augmentation is applied to rotate, flip and zoom the image to make the model strong and reduce the risk of overfitting.

3.1.3 Pre-Trained EfficientNet

EfficientNet is a powerful deep learning model which is which is designed to achieve high accuracy. So instead of training the model from the scratch we use transfer learning, which is employed by using a version of EfficientNet that has been pre-trained.

3.1.4 Model Training.

During the training phase the model learns to identify different visual features of different cattle breeds from the image. These features include various characteristics such as body size, skin texture, horn shape. The model is trained using the Adam optimizer and cross-entropy with label smoothing to improve classification accuracy and to reduce over fitting.

3.1.5 Model Saving.

The trained model is later saved in the .pth file format for the future use by the streamlit framework and converts the model into a web application where the user can access the site and predict the breed of an uploaded cattle image.

IV.EXPERIMENTAL SETUP AND SYSTEM IMPLEMENTATION.

4.1 Experimental Setup.

A. Dataset and Training Configuration.

The dataset consists of multiple cattle images these images were organize in separate class folders inside the main folder named dataset, this folder was later compressed into .zip file format. Later the dataset was split automatically into training and validation sets using a 80:20 ratio to ensure a proper model evaluation and generalization.

The images were resized to 224x224 pixel to match the input requirements of the EfficientNet architecture. Later the ImageNet normalization, to improve convergence during training. To enhance model robustness and reduce overfitting, Data augmentation techniques such as random rotation, horizontal flipping were applied to improve model robustness and reduce overfitting.

The model was trained using a batch size of 16 for 35 epochs using the Adam optimizer and cross-entropy loss Training was performed using Google Colab with GPU acceleration to ensure faster computation and efficient model optimization.

B. Model Training Procedure

The proposed livestock breed classification system is developed using a transfer learning approach based on the EfficientNetV2-S architecture pretrained on the ImageNet

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dataset. The pretrained network serves as a feature extractor capable of capturing general visual patterns such as edges, shapes, and textures. The custom classification head was added so that the model could adapt for livestock breed classification. During the training process, the network was made to discriminative features such as structure of the horn, face patterns, color of the coat and the body morphology. The model optimization was done using the Adam optimizer with the cross-entropy as the loss function to make sure the model learns effectively.

To improve the generalization and to reduce the overfitting, the standard data augmentation techniques ware used along with the normalization during the preprocessing and the dropout helps in stabilizing the training process. Later the training process is further enhanced by using the early stopping mechanism, where the validation performance is monitored and the risk of over fitting is prevented by the early stopping mechanism, where the model training is stopped when there is no further improvement.

The overall training strategy enables the model to achieve a good stability and a reliable classification performance across multiple livestock breeds.

4.2 Web Application Implementation.

The main idea of implementation of web application is to provide a user-friendly interface for real-time breed classification. The application was developed using the Streamlit Framework. The Streamlit was selected due to its simplicity, rapid deployment capability, and seamless integration with Python-based deep learning models.

A. Framework Selection.

Streamlit allows the development of interactive web application with the minimal frontend development. It supports the direct integration with PyTorch model, this makes it suitable for deploying the trained Deep Learning models. The framework allows the user to upload images and receive instant prediction with a lightweight and responsive user interface.

B. Model and Label Integration.

The trained model was exported in PyTorch format(.pth) as *indian_livestock_model-mix.pth*. This file contains the complete architecture and learned weights of the EfficientNet-based classification system. Additionally, breed class labels were stored in a JSON file (class_names-mix.json). This file preserves the mapping between output indices and corresponding breed names, ensuring consistent and accurate prediction labelling during deployment.

C. Prediction Workflow and Output Display.

Once the uploaded image preprocessing is completed, the image is passed to the trained EfficientNet -based model. The output layer generates probability scores for each breed class. The class with the highest probability is selected as the predicted breed. The associated probability value is converted into a percentage to represent the confidence level of the prediction. During the display of the output, the model displays the image that I uploaded by the user, the breed's name that has been predicted by the model with the confidence percentage and the top 3 prediction of the other breeds that look similar to the predicted breed and also their confidence score. This process allows the target user to easily interpret classification results without the need of any expertise knowledge.

V. PERFORMANCE EVALUATION AND RESULTS.

5.1 Model Performance Evaluation.

The overall performance of the proposed model that used EfficientNet and transfer learning for the breed detection was evaluated using the key metrics such as confusion metrics and accuracy of the model to ensure the reliable breed prediction results. As the task involved the process of breed recognition, both the measures such as the overall performance and the class-wise performance were analyzed during the model's performance evaluation.

A. Training and Validation Accuracy.

During the model's training process, the classification accuracy was used as the primary evaluation metric. A steady convergence was demonstrated by the model across the epochs, which improved the training accuracy and the validation accuracy progressively. This process showed that the model learned the features from the dataset and also the model maintained good performance on the unseen data.

B. Confusion Matrix Analysis.

To analyse the model's behaviour at different class-wise prediction, the confusion matrix was generated using the validation data. The confusion matrix helps to understand between the correctly and incorrectly classified samples across all the different breeds, by providing a detailed representation of these samples. The confusion matrix has two parts the diagonal elements and the off-diagonal elements, the diagonal elements represent the correctly classified entities. While the off-diagonal elements represent misclassifications. The analysis of the confusion matrix revealed that the breeds which are visually different achieved a higher classification accuracy and the other breed that has similar morphological features were showing a minor confusion as shown in the below figures.

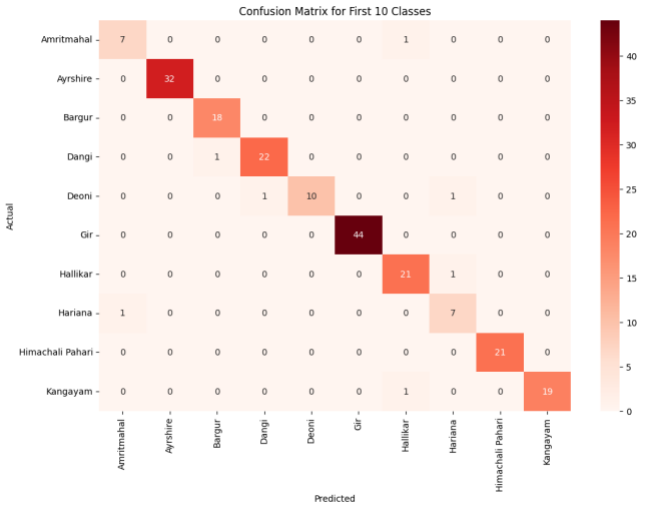

Figure 2. Confusion Matrix for Class 1-10

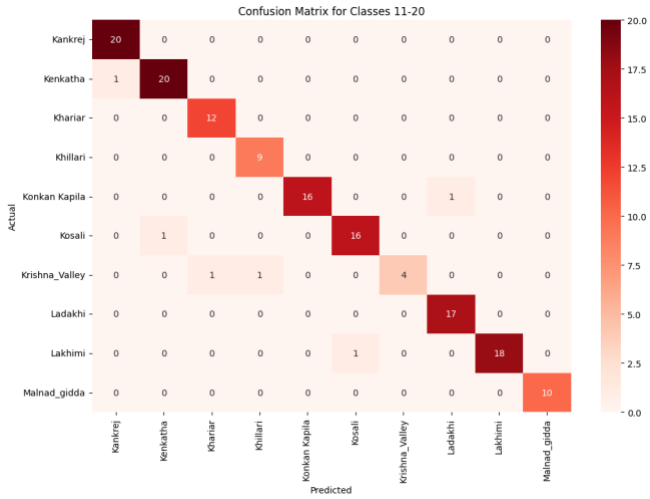

Figure 3. Confusion Matrix for Class 11-20

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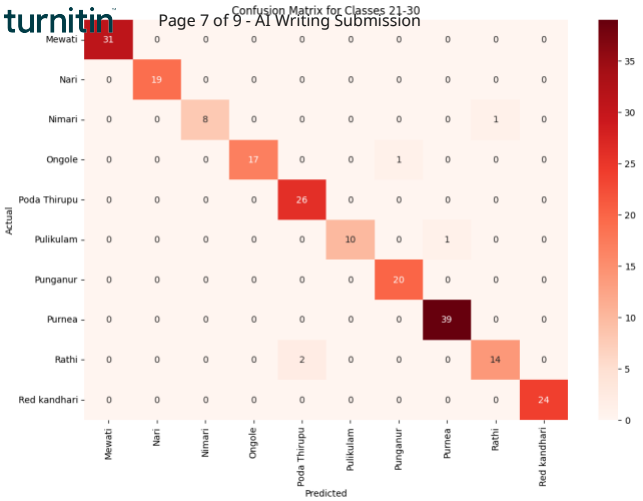

Figure 4. Confusion Matrix for Class 21-30

5.2 Web Application Performance.

Once the model is successfully trained and validated, the Indian livestock breed detection system was converted into a web application using the Streamlit framework, for the use of the non-technical users. The main idea of this application deployment was to validate the model's usability and the real-time breed prediction. The below figure 5 shows the web application interface.

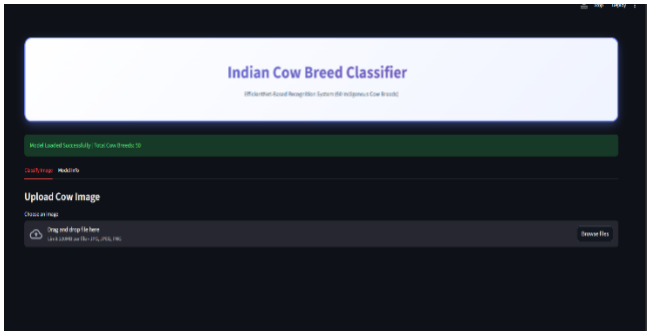

Figure 5. User Interface

A. Functional Validation.

The new phase of testing was conducted on the web application, where the application was tested by using the unseen images in order to assess the model's real-time performance. The user uploads an image, the system performs the preprocessing and visualizes the results without any errors or delays

The web application displays the breed's name present in the image along with the confidence percentage and the top-3 prediction that look similar to the predicted breed. This show the successful integration between the model that has been trained and the developed web application.

B. Prediction Efficiency.

The system that had been deployed as a web application demonstrated good efficiency, that contributed to the real-world predictions. The lightweight design of the system makes sure that the model is efficient computationally and also feasible for the practical deployment in real-world scenarios.

C. Usability and Accessibility.

The user interface is simple and easy to use; this allows the non-technical users such as farmers and livestock handlers to user the system without any pre-required knowledge. The image upload option and the visualization of the predicted results help in easy accessibility and the real-world application. The below figure 6 shows the prediction result for an uploaded image with the confidence score along with the top-3 predictions. The image of Bargur cow was uploaded and the model correctly identifies it as Bargur cow with a confidence score of 92.03% along with the top three predictions as Bargur, Nimari and Rathi respectively with their scores.

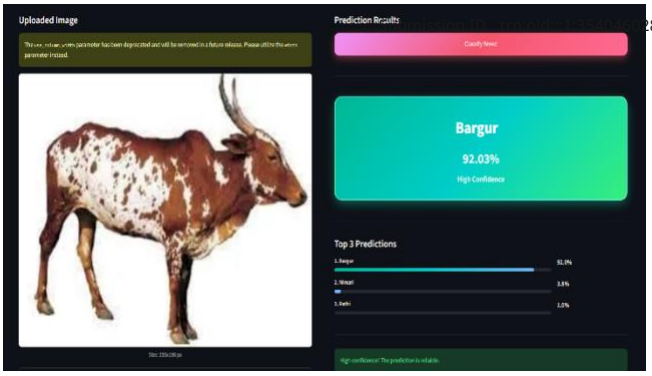

Figure 6. Breed Prediction Result

D. Deployment Reliability.

The multiple testing of the application, confirmed the stability and performance of the application across different images. The process of integrating the trained model which was save in .pth format along with the class names stored in a .json file ensures the consistent mapping accurately between the predicted indices and the breed names that has been stored.

This successful implementation of the web application shows that the proposed idea of turning a deep learning model into an application is not only technically effective but can also be implemented practically in the real-world livestock management.

V.CONCLUSION.

This paper presented the use of a deep learning framework in the field of livestock breed recognition system; which is capable of identifying multiple cattle breeds from the images. The idea of using a transfer learning approach along with the EfficientNet architecture helped in extracting multiple features from the images. The model training was performed by training the model on a dataset consisting of multiple cattle images related to different breeds, belonging to various regions of the country. The idea of using the techniques such as Image-preprocessing, data augmentation and regularization helped in improving the generalization of the model and to reduce the risk of overfitting.

The evaluation of the system was done based on the accuracy of the model and the analysis of the confusion matrix that demonstrated reliable performance of the model. These results indicated how strong the model discriminated between the visually distinct breeds and of the breeds that had similar features a slight misclassification was observed.

To enhance the model's practical usability, the model that had been trained was implemented as a web application using the Streamlit framework. The web application helped in real-time image upload by the users and the instant breed prediction for the uploaded image along with the confidence score and the top-3 prediction for the uploaded image.This made the system more accessible to the farmer, researchers and livestock management stakeholders so that the breed prediction process could be carried out without the need of manual experts or any physical tagging

Overall, the proposed system "EfficientNet-Based Transfer Learning Framework for Indian Cattle Breed Recognition" demonstrated technical effectiveness as well as real world application, by providing a scalable and an efficient solution for the process of automated livestock breed identification.

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